



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 7

1. **Aim:** To perform programming exercise on Hive.
2. **Objectives:**
 - To understand Program applications using tools like Hive.
 - To implement Hive programming model.
3. **Outcomes:** After study of this experiment, the student will be able to
 - Perform hands on Hive programming models.
4. **Prerequisite:** Core Java, Database concepts of SQL, Hadoop File system, and any of Linux operating system flavors.
5. **Requirements:** PC, Internet and VMWare software, Cloudera.

6. Pre-Experiment Exercise:

Brief Theory:

Students need to write the theory of following points.

Hive

1. Introduction of Hive.
2. Features of Hive.
3. Architecture of Hive.
4. Working of Hive with proper stepwise explanation.

7. Laboratory Exercise

A. Procedure

Open VMWare Workstation > select Cloudera > open virtual machine > in terminals

Perform the Hive commands.

(Minimum 20 commands should be executed)

Commands:

DDL Commands in Hive

1. DDL Commands in Hive
2. Drop Database in Hive
3. Describe Database Command in Hive
4. Alter Database Command in Hive
5. Show Database Command in Hive
6. Use Database Command in Hive
7. DDL Commands on Tables in Hive
8. Create Table Command in Hive



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9. DROP Table Command in Hive
10. TRUNCATE Table Command in Hive
11. ALTER Table Command in Hive
12. DESCRIBE Table Command in Hive
13. Show Table Command in Hive

Screenshots:

DDL Commands on Databases

1. Enter hive shell

The screenshot shows a terminal window titled "training@localhost:~". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content shows the command "[training@localhost ~]\$ hive" being entered, followed by the output "Hive history file=/tmp/training/hive_job_log_training_202504070120_59137372.txt" and the prompt "hive>". The window's title bar and the bottom status bar are visible. The status bar shows "training@localhost:~", "Applications", "Places", "System", and the time "1:20 AM".

2. Create database



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```
training@localhost:~
File Edit View Terminal Tabs Help
[training@localhost ~]$ hive
Hive history file=/tmp/training/hive_job_log_training_202504070120_59137372.
txt
hive> CREATE DATABASE sample;
OK
Time taken: 1.388 seconds
hive>
```

3. Show all databases



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```
training@localhost:~  
File Edit View Terminal Tabs Help  
txt  
hive> CREATE DATABASE sample;  
OK  
Time taken: 1.388 seconds  
hive> SHOW DATABASES;  
OK  
bda  
books  
data1  
db  
default  
employee  
example  
kriti  
mobilox  
mom  
sample  
stud  
test  
Time taken: 0.115 seconds  
hive>
```

4. Use database

```
Time taken: 0.115 seconds  
hive> USE sample;  
OK  
Time taken: 0.018 seconds  
hive>
```

5. Describe Database

```
hive> DESCRIBE DATABASE sample;  
OK  
sample          hdfs://localhost/user/hive/warehouse/sample.db  
Time taken: 0.026 seconds  
hive>
```

6. Alter Database

```
Time taken: 0.026 seconds  
hive> ALTER DATABASE sample SET DBPROPERTIES ('edited-by'='Eevee');  
OK  
Time taken: 0.046 seconds  
hive>
```



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7. Drop Database

```
hive> DROP DATABASE bda;  
OK  
Time taken: 0.052 seconds  
hive>
```

DDL Commands on Tables

1. Create Table

```
hive> CREATE TABLE employee(id INT, name STRING, salary FLOAT);  
OK  
Time taken: 0.094 seconds  
hive>
```

2. Show Tables

```
hive> SHOW TABLES;  
OK  
employee  
Time taken: 0.03 seconds  
hive>
```

3. Describe Table

```
hive> DESCRIBE employee;  
OK  
id      int  
name    string  
salary  float  
Time taken: 0.042 seconds  
hive>
```

4. Alter Table

```
hive> ALTER TABLE employee ADD COLUMNS (age INT);  
OK  
Time taken: 0.045 seconds  
hive> DESCRIBE employee;  
OK  
id      int  
name    string  
salary  float  
age     int  
Time taken: 0.032 seconds  
hive>
```

5. Drop Table



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```
hive> DROP TABLE employee;  
OK  
Time taken: 0.138 seconds  
hive> █
```

Conclusion:

Thus we successfully implemented DDL Commands using Hive.